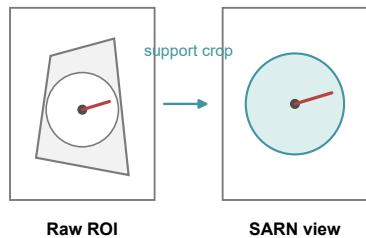
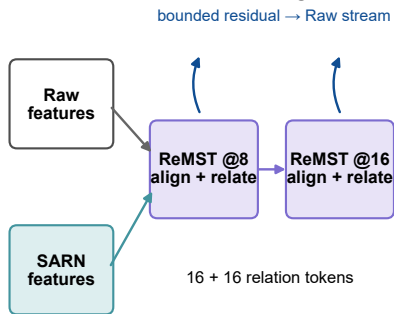


a Paired observations

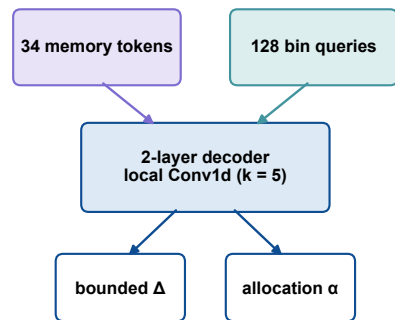


b Hierarchical relation encoding



shared frozen encoder; relation modules train

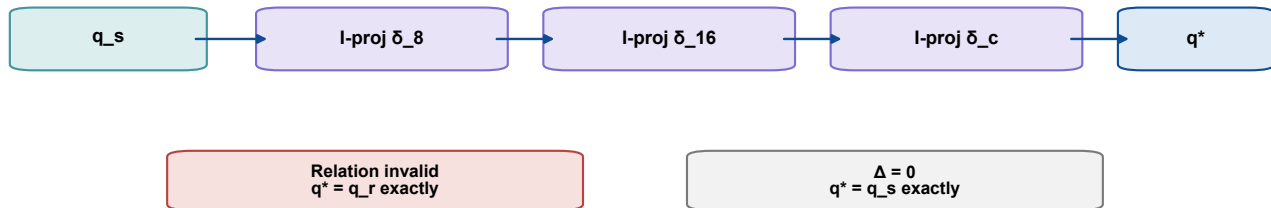
c Progress-aware coordination



d

Feasible moment transport and explicit boundary

Requested shifts sum to Δ ; targets clip to $[\text{eps}, 1 - \text{eps}]$
clipping may shorten movement but cannot reverse it



Output: $E[q^*]$